

Ujaan Das

+44-74350-84812 | ujaandas03@gmail.com | [linkedin.com/in/ujaandas](https://www.linkedin.com/in/ujaandas) | github.com/ujaandas

EDUCATION

University of Warwick

Master of Science in Computer Science (Predicted Distinction)

Hong Kong University of Science and Technology

Bachelor of Engineering in Computer Engineering

Coventry, United Kingdom

Sep. 2025 – Sep. 2026

Hong Kong SAR

Sep. 2021 – May 2025

EXPERIENCE

Stellarus Technology

Software Engineering Intern (Part Time)

Jan. 2025 – Aug. 2025

Hong Kong SAR

- Re-designed a chatbot execution pipeline using a multi-step LLM workflow, implementing a custom command DSL for structured tool orchestration that reduced token overhead by 25% per session
- Standardized message contracts and schema enforcement across the pipeline, eliminating 90% of parsing-related regressions and reducing runtime errors by 30%
- Integrated into SpaceGPT and awarded a Gold Medal at the 50th Geneva International Invention Exhibition

Stellarus Technology

Software Engineering Intern

May 2023 – Aug. 2023

Hong Kong SAR

- Developed RESTful APIs using FastAPI and MongoDB for multi-user task queues for live satellite data queries
- Designed to sustain throughput of 8,000+ requests/sec with <150ms latency and minimal queue times
- Parallelized CPU-bottlenecked operations via a Celery/Redis worker pool, reducing processing time by 34%
- Partnered with frontend team to containerize the SPA and backend using Docker and orchestrated automated CI/CD pipelines for Azure Container Apps, cutting deployment time by 20%

HCI Initiative, HKUST

Research Assistant

Jun. 2022 – Aug. 2022

Hong Kong SAR

- Developed a React/TypeScript SPA utilizing Custom Hooks and Context API for centralized state handling
- Engineered a real-time telemetry and metrics pipeline processing 50K+ events/session, enabling analysis across 4K+ user interactions in large-scale experiments
- Worked with other post-docs to build a standardized TailwindCSS and MaterialUI component library
- Contributed to FARPLS, a feature-augmented preference learning system (published at ACM/IUI'24)

PROJECTS

MSc Dissertation - Invariants | C++, Google Test Suite, CMake, Python

[Github](#)

- Applying formal systems theory to LLMs for improved structured output generation via runtime constraints
- Architecting a C++ DSL and compiler with Python bindings for seamless integration into inference pipelines

Snip | C++, Termios, Google Test Suite, CMake, Nix

[Github](#)

- Actively maintaining a modal text editor built from ground-up with asynchronous, event-driven architecture

TerraViz | Next.js, React, TypeScript, CesiumJS, TailwindCSS

[DevPost](#)

- Built an interactive geospatial platform in Next.js/React for real-time exploration of climate and land-use data

oojsite | Go, HTML, TailwindCSS, Nix

[Github](#)

- Built a static site generator in Go to compile Markdown into composable HTML, actively used for [my blog](#)

SKILLS

Languages: C++, Python, Go, C, TypeScript, SQL, JavaScript, HTML/CSS

Frameworks & Libraries: Google Test Suite, CMake, FastAPI, Next.js, React.js, Node.js, LangGraph, Pydantic

Databases: MongoDB, PostgreSQL, MySQL, SQLite, Redis

Tools & Cloud: Docker, Nix, Azure, AWS, GitHub Actions, NGINX, Make, Git

AWARDS

HackEurope'26 Scholar, Paris | *Selected for full scholarship by SIG, Anthropic, and Warwick*

Mar. 2026

Track Runner Up, Imperial College London ICHack | *Tackled HRT's Big Data challenge*

Feb. 2026

Track Winner, Cathay Pacific Hackathon | *Ranked 1st out of 1,000+ global applicants (Top 0.01%)*

Nov. 2024

Published Researcher, ACM IUI'24 | *Paper on human-AI interaction*

Mar. 2024

Global Merit Scholar | *Competitive scholarship for study abroad at Northwestern University, USA*

2023